

A Control Function Approach to Endogeneity

ECMT 676 (Econometric II) Lecture Notes

TAMU

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- Control Function Approaches to Endogeneity
- Testing for Endogeneity
- Testing Overidentifying Restrictions

Control function approaches to endogeneity

- Most models that are linear in parameters are estimated using standard IV methods – two stage least squares (2SLS).
- An alternative, the control function (CF) approach, relies on the same kinds of identification conditions.
- However, in models with nonlinearities or random coefficients, the form of exogeneity is stronger and more restrictions are imposed on the reduced forms.

- Let y_1 be the response variable, y_2 the endogenous explanatory variable (EEV), and $\mathbf{z}$ the $1 \times L$ vector of exogenous variables (with $z_1 = 1$):

$$y_1 = \mathbf{z}_1 \delta_1 + \alpha_1 y_2 + u_1, \quad ((1))$$

where $\mathbf{z}_1$ is a $1 \times L_1$ strict subvector of $\mathbf{z}$.



$$\overbrace{(z_1, z_2, \dots, z_{L_1}, z_{L_1+1}, \dots, z_{L-1}, z_L)}^{=\mathbf{z}}$$

$$\underbrace{\hspace{1.5cm}}_{\mathbf{z}_1} \quad \underbrace{\hspace{1.5cm}}_{\text{excluded IVs}}$$

- Reduced form for y_2 :

$$y_2 = \mathbf{z} \pi_2 + v_2, \quad E(\mathbf{z}' v_2) = \mathbf{0} \quad ((3))$$

where π_2 is $L \times 1$.

- Consider the (weakest) exogeneity assumption $E(\mathbf{z}' u_1) = \mathbf{0}$.

- Write the linear projection of u_1 on v_2 , in error form, as

$$u_1 = \rho_1 v_2 + e_1, \quad ((4))$$

where

$$\rho_1 = E(v_2 u_1) / E(v_2^2)$$

is the population regression coefficient.

- By construction, $E(v_2 e_1) = 0$ and $E(\mathbf{z}' e_1) = \mathbf{0}$.
- Endogeneity is now fully reflected in $E(u_1 v_2)$ or in ρ_1 , since from (3) we have

$$\text{cov}(u_1, y_2) = \underbrace{\text{cov}(u_1, \mathbf{z}\pi_2)}_{=0} + \text{cov}(u_1, v_2)$$

- We then have

$$y_1 = \mathbf{z}_1 \delta_1 + \alpha_1 y_2 + \underbrace{\rho_1 v_2 + e_1}_{=u_1}, \quad ((5))$$

where v_2 is now an explanatory variable in the equation.

- The new error, e_1 , is uncorrelated with y_2 as well as with v_2 and $\mathbf{z}$.

- Two-step procedure:

- ▶ (i) Regress y_{i2} on $\mathbf{z}_i$ and obtain the reduced form residuals, $\hat{v}_{i2}$;
- ▶ (ii) Regress

$$y_{i1} \text{ on } \mathbf{z}_{i1}, y_{i2}, \text{ and } \hat{v}_{i2}. \quad ((6))$$

- The implicit error in (6) is $e_{i1} + \rho_1 \mathbf{z}_i (\hat{\pi}_2 - \pi_2)$, which depends on the sampling error in $\hat{\pi}_2$ unless $\rho_1 = 0$ (exogeneity test).
- OLS estimators from (6) will be consistent for δ_1, α_1 , and ρ_1 .
- The OLS estimates from (6) are control function estimates of (δ_1, α_1) .

- Algebra: The OLS estimates of δ_1 and α_1 from (6) are *identical* to the 2SLS estimates of

$$y_1 = \mathbf{z}_1 \delta_1 + \alpha_1 y_2 + u_1$$

- Can prove this using Frisch-Waugh-Lovell (FWL): Define $\mathbf{X}_1 = (\mathbf{Z}_1, \mathbf{Y}_2)$.
- OLS coefficients $\hat{\beta}_1$ on $\mathbf{X}_1$ can be obtained as

(i) Regress $\mathbf{X}_1$ on $\hat{\mathbf{V}}_2$, get residuals, say $\tilde{\mathbf{X}}_1$.

(ii) Regress $\mathbf{Y}_1$ on $\tilde{\mathbf{X}}_1$ to get $\hat{\beta}_1$. (here we use FWL: such obtained $\hat{\beta}_1$ is identical to the CF estimate)

- Result follows by showing $\underset{\text{CF}}{\tilde{\mathbf{X}}_1} = (\underset{\text{2SLS}}{\mathbf{Z}_1}, \hat{\mathbf{Y}}_2)$.

Testing for Endogeneity

- A regression-based Hausman test uses the control function approach. Write

$$y_1 = \mathbf{z}_1 \delta_1 + \mathbf{y}_2 \alpha_1 + u_1, \quad ((28))$$

where $\mathbf{z}_1$ is $1 \times L_1$, $\mathbf{y}_2$ is $1 \times G_1$, and the entire vector of all instruments is $\mathbf{z} = (\mathbf{z}_1, \mathbf{z}_2)$, where $\mathbf{z}_2$ is $1 \times L_2$ with $L_2 \geq G_1$.

- Write the reduced forms as

$$\begin{aligned} \mathbf{y}_2 &= \mathbf{z} \Pi_2 + \mathbf{v}_2 \\ E(\mathbf{z}' \mathbf{v}_2) &= \mathbf{0} \end{aligned}$$

- For a $G_1 \times 1$ vector $\boldsymbol{\rho}_1$ write the linear projection of u_1 on $\mathbf{v}_2$:

$$\begin{aligned}u_1 &= \mathbf{v}_2 \boldsymbol{\rho}_1 + e_1 \\ E(\mathbf{v}_2' e_1) &= \mathbf{0}.\end{aligned}$$

We also know $E(\mathbf{z}' e_1) = \mathbf{0}$.

- Writing

$$y_1 = \mathbf{z}_1 \boldsymbol{\delta}_1 + \mathbf{y}_2 \boldsymbol{\alpha}_1 + \mathbf{v}_2 \boldsymbol{\rho}_1 + e_1$$

leads to a simple two-step procedure.

- (i) Regress $\mathbf{y}_{i2}$ on $\mathbf{z}_i$ to obtain the $1 \times G_1$ reduced form residuals, $\hat{\mathbf{v}}_{i2}$ (one vector for each observation i).

This can be done for each element of $\mathbf{y}_{i2}$ separately.

- (ii) Run the regression

$$y_{i1} \text{ on } \mathbf{z}_{i1}, \mathbf{y}_{i2}, \hat{\mathbf{v}}_{i2} \quad ((29))$$

and use a joint Wald test of $H_0 : \boldsymbol{\rho}_1 = \mathbf{0}$, where $\boldsymbol{\rho}_1$ is the vector of coefficients on $\hat{\mathbf{v}}_{i2}$.

- (This is often computed as an approximate F statistic by dividing the Wald statistic by G_1 , the number of restrictions being tested.)

Testing Overidentifying Restrictions

- Hausman test is useful to test y_2 is endogenous
- Now we aim to test z is exogenous
- We focus on the regression-based version of both tests

If we have more instruments than we need we can, in a (weak) sense, test whether some of them are exogenous.

Write the equation as

$$y_1 = \mathbf{z}_1 \delta_1 + \mathbf{y}_2 \alpha_1 + u_1 \quad ((32))$$

where $\mathbf{z}_1$ is $1 \times L_1$ and $\mathbf{y}_2$ is $1 \times G_1$.

The entire vector of instruments is $\mathbf{z} = (\mathbf{z}_1, \mathbf{z}_2)$, where $\mathbf{z}_2$ is $1 \times L_2$.

The equation is overidentified if $L_2 > G_1$.

- The 2SLS estimator uses $L_1 + L_2$ moment conditions to estimate $L_1 + G_1$ parameters, so $L_2 - G_1$ overidentifying restrictions can be tested.

- As with testing for endogeneity, regression-based tests are convenient. Under homoskedasticity (Assumption 2SLS.3), obtain NR_u^2 (generally, the uncentered R -squared, but almost always the usual R -squared) from

$$\hat{u}_{i1} \text{ on } \mathbf{z}_i, \quad ((33))$$

where $\hat{u}_{i1}$ are the 2SLS residuals and $\mathbf{z}$ is the vector of all exogenous variables.

- The motivation for (33) is that, under the null, we should have, in the sample,

$$N^{-1} \sum_{i=1}^N \mathbf{z}'_i \hat{u}_{i1} \approx \mathbf{0}. \quad ((34))$$

- There are not as many degrees-of-freedom as (34) might seem to suggest.
- Under the null hypothesis

$$E(\mathbf{z}'u) = \mathbf{0} \quad ((36))$$

$$E(u^2 \mathbf{z}'\mathbf{z}) = \sigma^2 E(\mathbf{z}'\mathbf{z}) \quad ((37))$$

it can be shown

$$NR_u^2 \overset{a}{\sim} \chi^2_{L_2 - G_1}. \quad ((38))$$

- Easy to compute, but not robust to heteroskedasticity.
- The test has the wrong asymptotic size if (37) fails, but the test has no systematic power for detecting failure of (37).